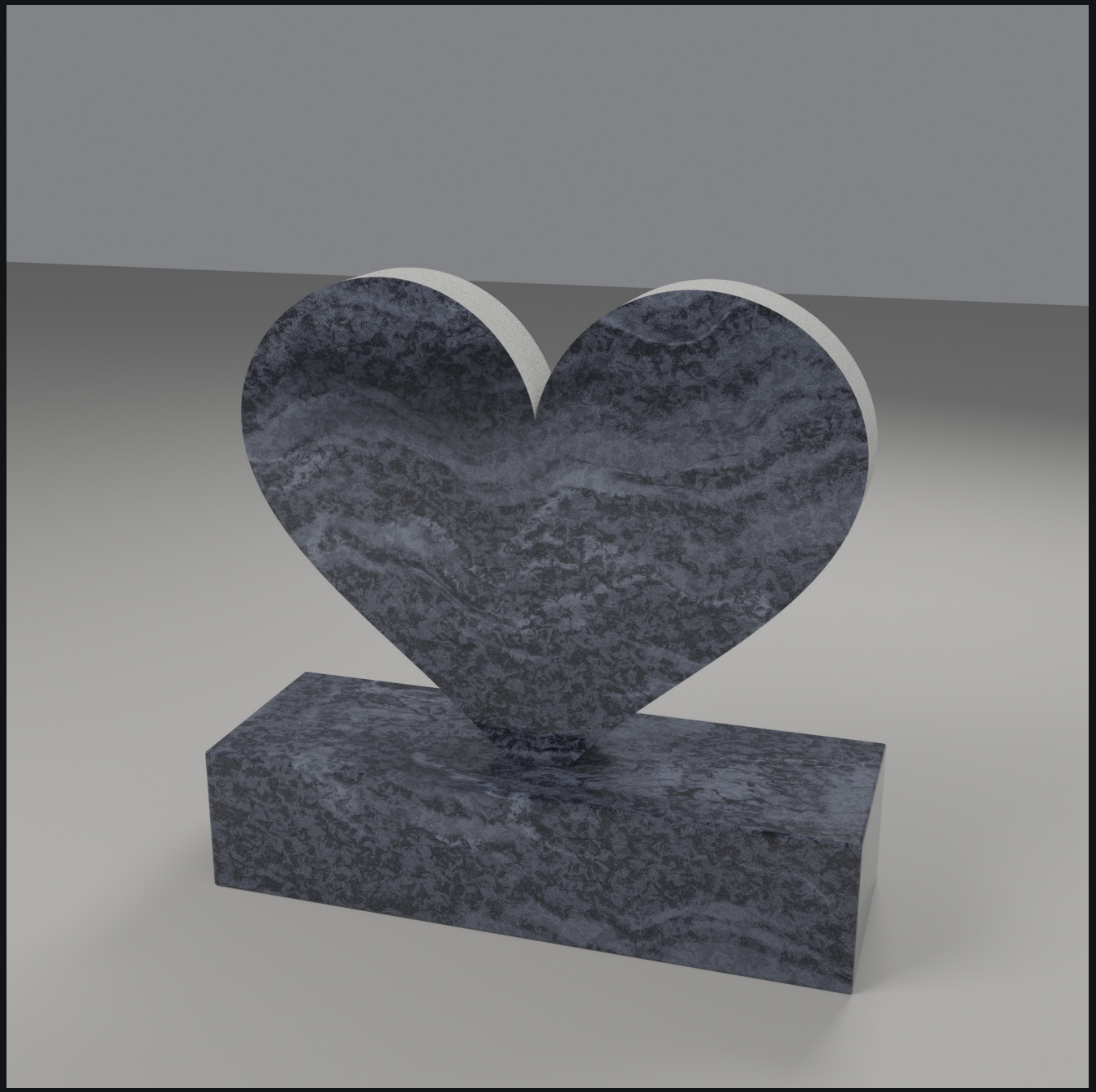
MATERIAL REPORT · DATASET "ADDITIONAL"

Bahama Blue

GS07_BahamaBlue

A seed-controlled Blender material for blue-black wavy gneiss (the "BHB" stone of the heart memorials and PARVA / ALLEVO layouts). It was calibrated against a 1 m reference swatch and the workshop photos.



FINISHES
Polished · Flamed · Honed

REFERENCE
1 m swatch at 0.49 mm/px + 2 photos

SEED CONTROL
material, per object, automatic

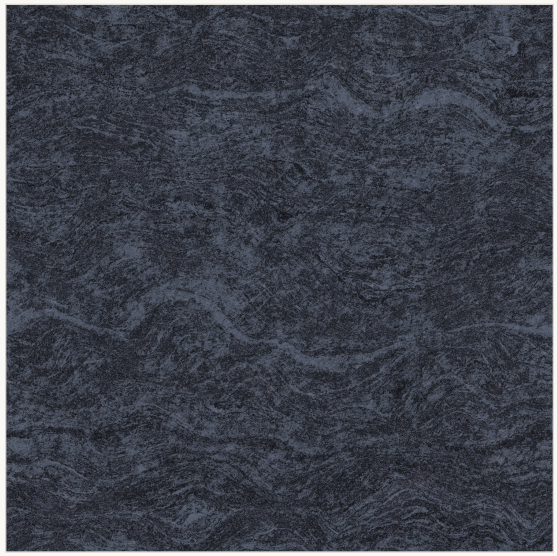
TESTED
Blender 5.2 LTS · Cycles

October 2026

REFERENCE STUDY

What the reference shows

additional folder: 5 images



TEXTURE SWATCH bhb3--less-wavy · 1 m · 2048 px

PARVA SLAB BHB_PARVA render

HEART, POLISHED FACE workshop photo

HEART, FLAMED RIM workshop photo

Polished face

- Blue-black base, sRGB about 56 60 71. Blue exceeds red by about 15, and there are no red or rose minerals.
- Near-horizontal layers that bend together in soft waves about 25–35 cm long and 3–5 cm high.
- Up close: a lace-like dark network of fine crystals (cells about 1–2 cm) over a slate-blue background.
- Thin pale wavy streaks and a few thin dark streaks follow the layering. Crystals are under 1 mm.

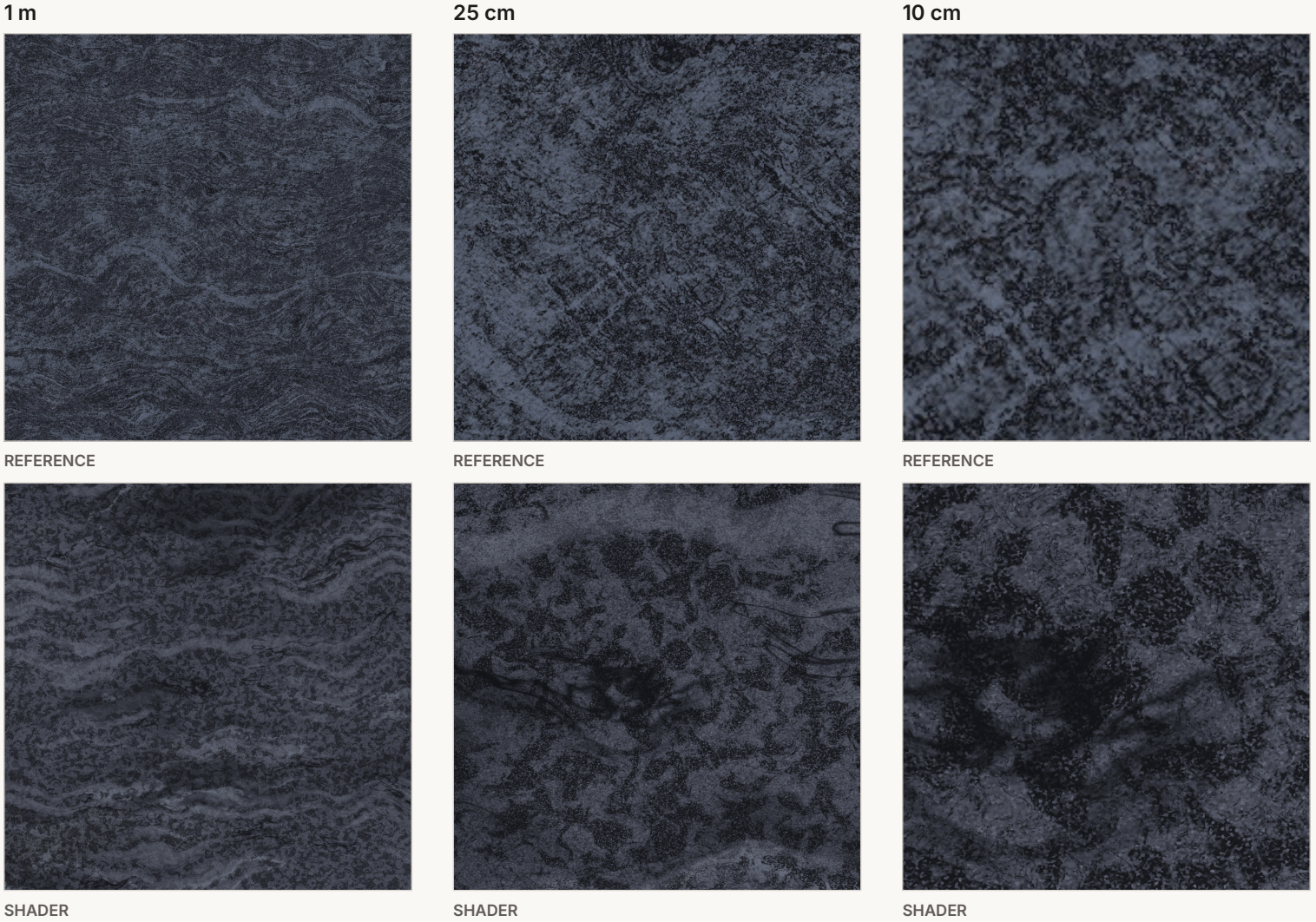
Edges and finish

- Heart rims and tops are flamed: a pale grey surface (sRGB about 187) with irregular dark grains 2–4 mm across.
- The flamed surface is rough with fine relief; the face and base are mirror-polished.
- The swatch’s two-direction spectrum shows a weak horizontal preference (anisotropy about 1.2–1.9), not a strong grain.

CALIBRATION

Reference vs shader at the same physical size

top: swatch crop · bottom: shader albedo, 2048 px



PATCH	REF MEAN SRGB	REF σ	REF P5–P95	SHADER MEAN	SHADER σ	SHADER P5–P95
1 m	56 60 71	15.9	38–90	**61 64 74**	**15.8**	**40–91**
25 cm	58 63 74	23.1	23–99	**55 58 67**	**17.3**	**29–86**
10 cm	57 62 72	22.3	25–97	**53 55 64**	**19.3**	**26–86**

The overall colour and the large-scale wave structure match closely at 1 m. At 10–25 cm the shader has about 20% less contrast and slightly duller light blue areas than the swatch.

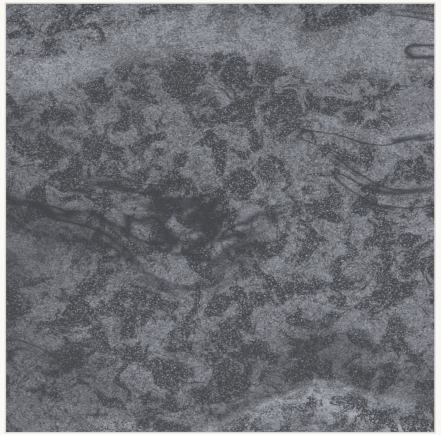
FINISHES

Flamed and honed surfaces

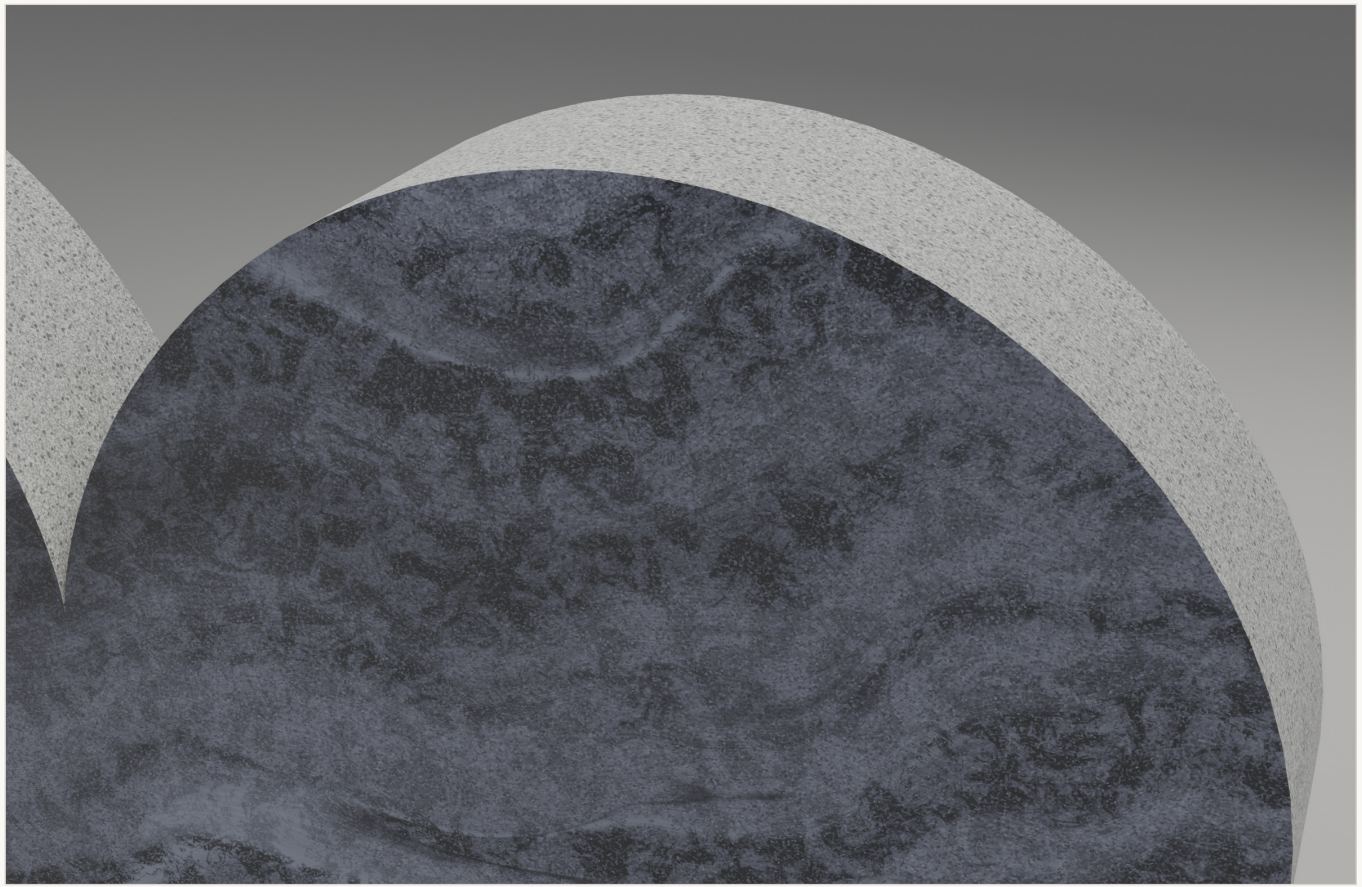
photo strip vs shader · lit close-up render

PHOTO flamed rim

SHADER FLAMED 10 cm



SHADER HONED 25 cm



LIT CLOSE-UP Cycles · 2400 × 1600 · polished face meets flamed rim

Flamed (rims, heart tops)

Every dark crystal is exposed as a discrete speck with a ragged outline on a pale ground, and the pale crystals vary one by one in brightness. Measured on the rim: photo mean sRGB 187 186 181, σ 31, p5–p95 131–231; shader 188 187 183, σ 26, 136–222. Uses a fine bump; no displacement needed.

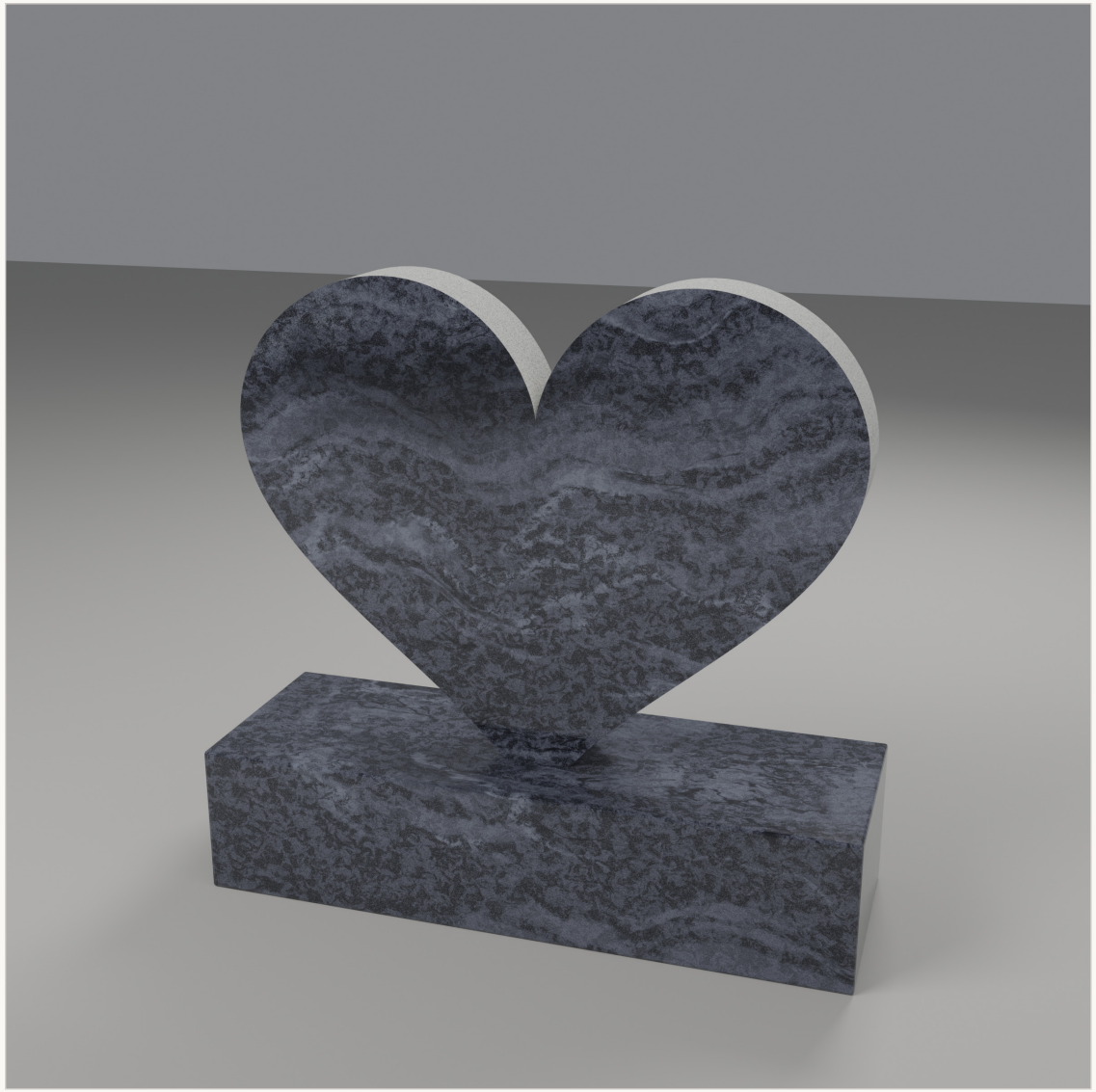
Honed (matte sides)

Honing lifts the slate-blue layers while the dark network stays dark, so the waves stay readable on matte faces. Used on the PARVA slab's sides. Roughness: polished 0.06, honed 0.55, flamed 0.85.

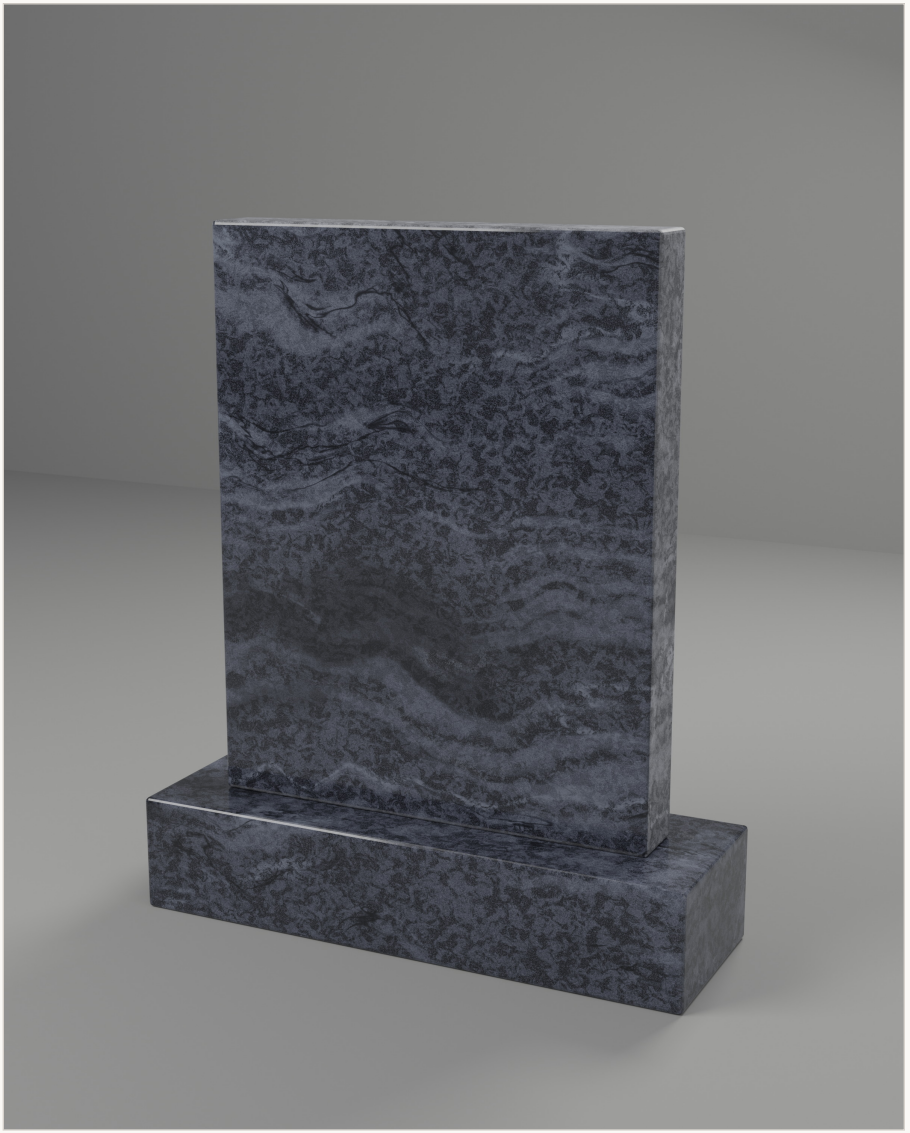
RENDERS

The material on memorials

Cycles · HD · default seed 7



HEART MEMORIAL 2400 × 2400 · polished face, flamed rim

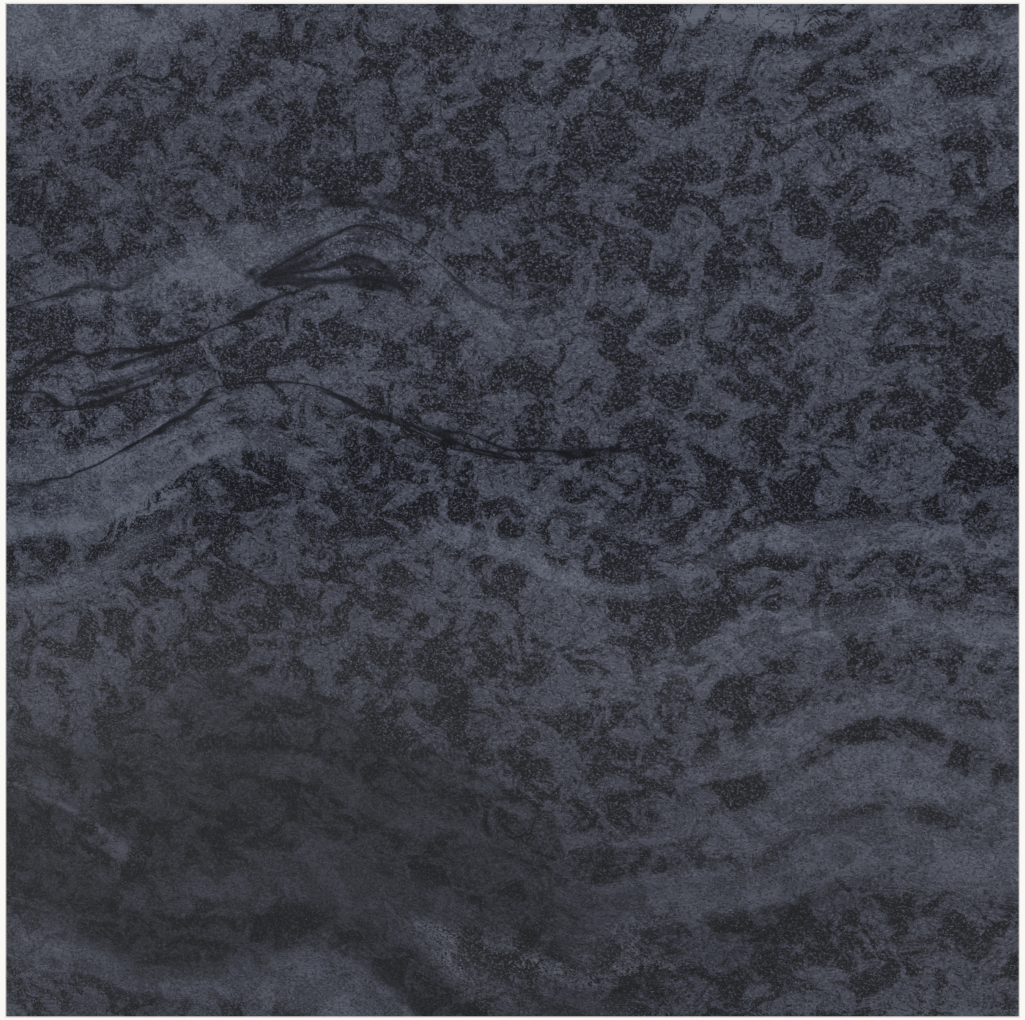


PARVA-STYLE SLAB 2400 × 3000 · polished, honed sides

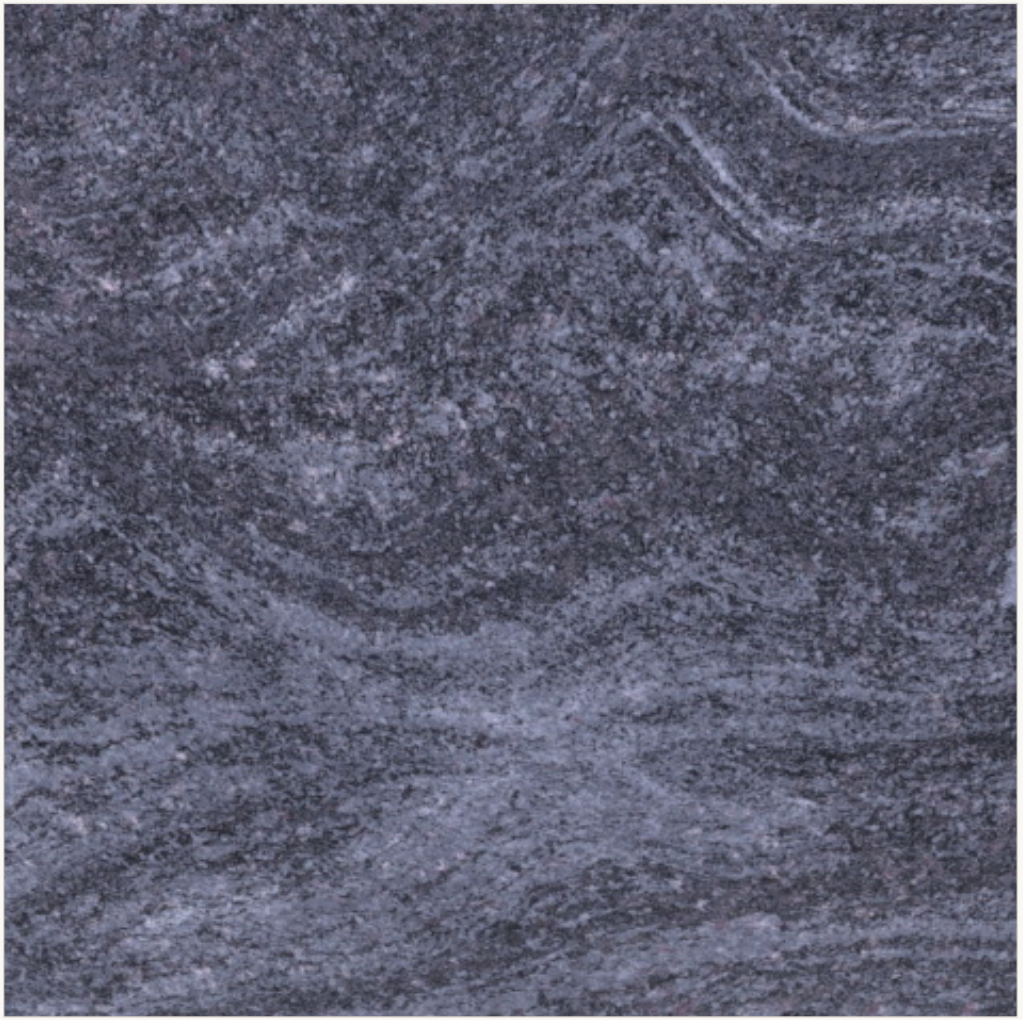
RENDERS

Face detail

lit close-up of the polished face · 2400 × 2400



SHADER Cycles render

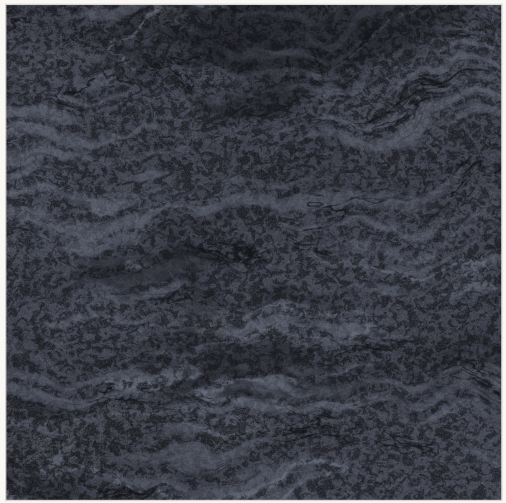


REFERENCE BHB_PARVA, similar area

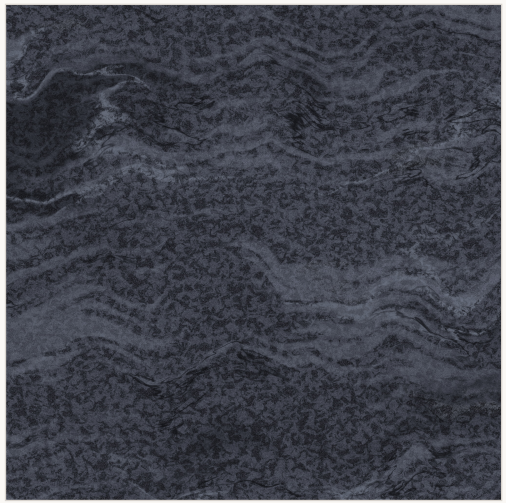
SEED CONTROL

One material, many stones

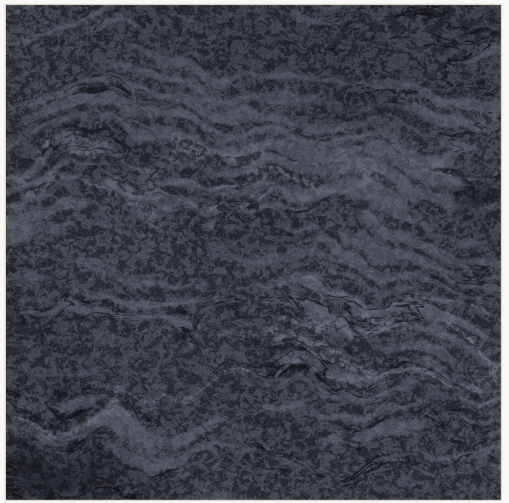
same preset, different seeds



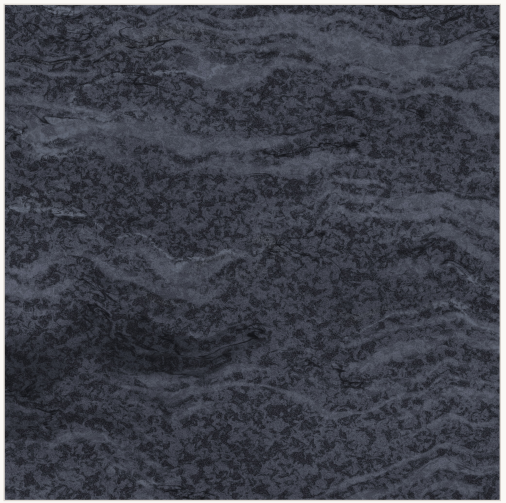
SEED 7 (DEFAULT) 1 m patch



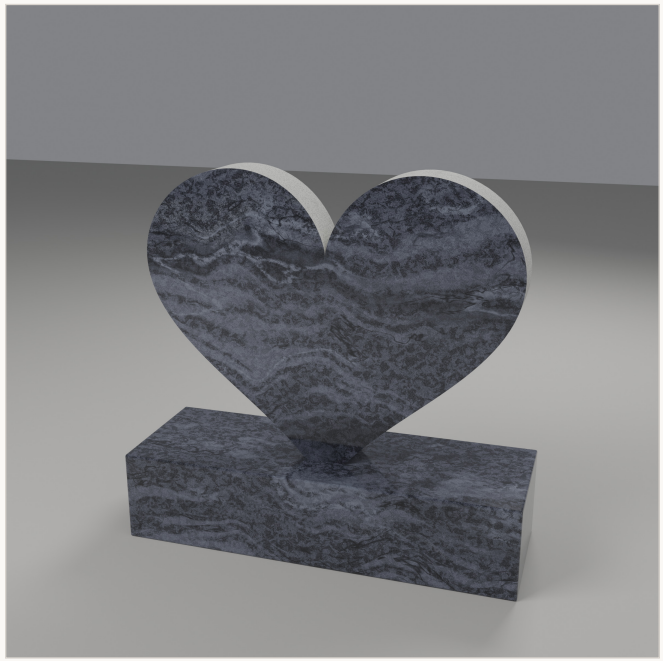
SEED 77 1 m patch



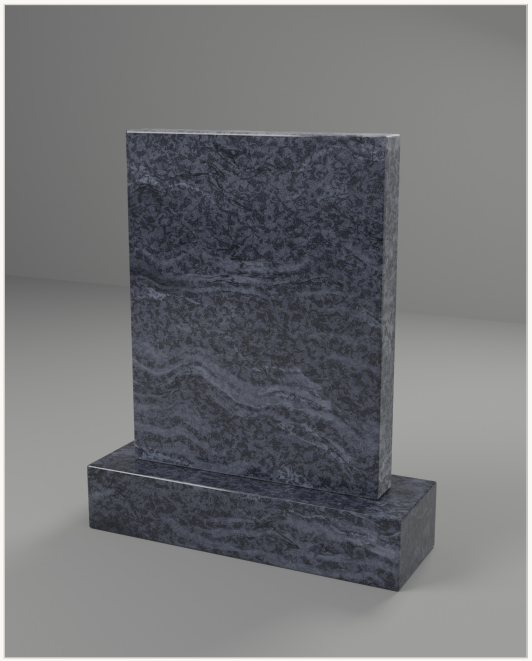
SEED 777 1 m patch



SEED 7777 1 m patch



HEART, SEED 2026



SLAB, SEED 2026

Three levels, added together

- Material: the Granite node’s Seed input, SEED in the script, or set_seed(42, "GS07_BahamaBlue").
- Per object: the granite_seed custom property, set_object_seed(obj, 7) or randomize_selected_seeds().
- Automatic: Object Random Seed = 1 makes duplicates differ; 0 makes them identical.
- Seed Variation (0–1) also nudges warp and layer amounts.

Verified: correlation between seed 7 and seeds 77 / 777 / 7777 = 0.005 / 0.015 / −0.006, so the patterns are unrelated; re-entering seed 7 gives 1.000.

PARAMETERS

Preset and new controls

additional/additional_GS07_BahamaBlue.py

New controls added for this stone

Undulation, Undulation Scale
bends the whole layer stack so layers wave together

Mottle, Mottle Scale, Mottle Threshold
the lace-like dark crystal network

Light Veins
thin pale wavy streaks along the layering

Flamed specks (preset)
speck size, density and ragged outline on flamed faces

These controls default to off, so the six earlier materials render the same as before (re-checked: 0.1% of pixels differ by more than 1/255, which is render noise).

How to use

- Blender 4.0+ → Scripting → open the script → Run Script. Selected meshes get GS07_BahamaBlue_Polished.
- Assign GS07_BahamaBlue_Flamed to rims and tops, and _Honed to sawn or matte faces.
- Object coordinates in metres, front face in the X/Z plane; apply scale (Ctrl+A).

Preset values

scale	5	grain scale	320
flow angle	0	light veins	0.6
stretch	0.3	mottle	0.95
warp	1	mottle scale	0.045
fold	0.5	mottle thr	0.5
crinkle	1.2	vein width	0.03
undulation	1.3	mottle porosity	0.5
undulation scale	2.7	brightness	0.9
rose	0.38	polish rough	0.06
melano	0.25	rock speckle	0.85
biotite	0.6	flame speck size	0.3
lamination	0.25	flame speck density	0.55
lam scale	60	flame speck ragged	0.4